

Use BLOCKCHAIN TECHNOLOGY To Rout Risk Out Of Network Transactions



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Ripple, a blockchain based technology startup, is sitting on a venture capital of US\$ 93 million, sourced from an interesting mix of funders ranging from Google Ventures and Seagate Technology to a horde of banks like Standard Chartered. Ripple's network already has 15 global banks, and 10 more are waiting to board the train. Shanghai Huarui Bank, a privately-owned bank in China, is working with Ripple to implement a commercial cross-border payment service, so customers can transfer money internationally in real time. The bank feels it will help Chinese families to send money to their children studying abroad.

What is Ripple's offering that promises such confidence in the network? What is this technology, which interests not just banks but Web giants and the tech industry as well? In two words, the answer is blockchain technology.

Bank investment in blockchain technology is expected to reach US\$ 400

million by 2019. This is because of the confidence that blockchain technology has the potential to ensure completely secure and tamper-proof transactions, in scenarios ranging from smart contracts and real-time international money transfer to contactless payments. Early blockchain systems had their share of problems like settlement lag and lack of privacy, but the current generation of startups is bent upon delivering enterprise-grade blockchain.

Demystifying the blockchain

You do not have to feel bad if you have already Googled for blockchain and not made head or tail of it. Even finance and tech experts take some time to understand this concept, and even more time to explain it confidently. As with most nascent technologies, it is difficult to differentiate the technology from its applications, and understand it for what it really is, what it is currently used for and what else it can do.

A blockchain is a decentralised, distributed public ledger that records, validates and secures transactions in a system. As is obvious from the name, a blockchain is made of several blocks. Each block contains encrypted data pertaining to a transaction, along with a hash that draws upon the previous block in the chain.

This way of connecting each cryptographically-hashed block with the previous one ensures that all data in the chain remains secure and unchanged. There is no single owner of any transaction—each transaction is authorised and backed by thousands of computers (called miners)—and so there is no single point that can be attacked or hacked.

That is what makes blockchain based transactions secure and tamper-proof. Using blockchain technology, it is possible to

Fig. 1: Blockchain technology ensures completely secure and tamper-proof transactions in various scenarios (Image courtesy: ToneTag)

